18/03/2025

Lecture - 4

First fundamental theorem of calculus:

Let f be a continuous function on $[a,b]$. For $x \in [a,b]$, define

$$F(x) := \int_a^x f(t)\,dt.$$

Then, F is differentiable on $[a,b]$ and $F'(x) = f(x)$ for all $x \in [a,b]$.

Separable ODEs:

An ordinary differential equation of the form

$$M(x) + N(y)\frac{dy}{dx} = 0 \quad \text{— ①}$$

is called a separable ODE, where M and N are functions of x and y, respectively.

Let $H_1(x)$ and $H_2(y)$ be any functions such that

$$H_1'(x) = M(x) \quad \text{and} \quad H_2'(y) = N(y) \quad \text{— ②}$$

Substituting (2) in (1), we obtain

$$H_1'(x) + H_2'(y)\,y' = 0. \quad \text{— ③}$$

Using chain rule, $\frac{d}{dx}(H_2(y)) = H_2'(y)\frac{dy}{dx}$. — ④

Here (4) in (3) leads to

$$\frac{d}{dx}\left(H_1(x) + H_2(y) \right) = 0 \quad (5)$$

Integration of (5) gives

$$\boxed{H_1(x) + H_2(y) = c}$$

where c is an arbitrary constant.

Question: Why do such functions ($H_1(x)$, $H_2(y)$) exist?

Justification of existence of $H_1(x)$ and $H_2(y)$:

Since M and N are continuous functions of x and y, respectively, on $[a,b]$. Define

$$H_1(x) = \int_a^x M(x)\,dx \quad \text{and} \quad H_2(y) = \int_a^y N(y)\,dy.$$

Using the first fundamental theorem of calculus, $H_1(x)$ and $H_2(y)$ are differentiable functions and $H_1'(x) = M(x)$ & $H_2'(y) = N(y)$.

Question: Why do we call it as separable ODEs?

Reason: Note that if we compare (1) with $y' = f(x,y)$ then $f(x,y) = -M(x)\left(\frac{1}{N(y)}\right)$.

Write $h(x) = -M(x)$ and $g(y) = \frac{1}{N(y)}$ with $N(y) \neq 0$.

Then,

$$\boxed{f(x,y) = h(x)\, g(y)}$$

This is why, these ODEs are called separable ODEs.

Examples of separable ODEs :

Find the solution to the initial value problem

$$\underbrace{\frac{dy}{dx} = \frac{y\cos x}{1+2y^2}}_{DE}, \quad \underbrace{y(0) = 1.}_{\text{Initial value}}$$

Assume $y(x) \neq 0$. Then,

$$\left(\frac{1+2y^2}{y}\right) dy = \cos x \, dx.$$

On integrating, we get

$$\ln|y| + y^2 = \sin x + c$$

If we choose, initial condition $y(0) = 1$.

Then we get $c = 1$. Hence, a particular solution to the IVP is

$$\boxed{\ln|y| + y^2 = \sin x + 1}$$

Remark : Note that, $y(x) = 0$ is a solution to the given DE, but it is not a solution to the given IVP.

<u>Remark</u>: In finding a one parameter family of solutions in the separation process, we assume that $N(y) \neq 0$. Therefore, some solutions may be lost in the formal separation process.

<u>Example</u>: Solve the DE $\quad y' = -2xy \quad$ — ⊛

<u>solution</u>: Separating the variables we get

$$2x + \frac{y'}{y} = 0, \qquad y(x) \neq 0$$

Integrating both sides we obtain

$$\ln|y| + x^2 = c_1 \qquad (\text{Implicit})$$

Thus, the solutions are

$$\boxed{y(x) = c\,e^{-x^2}} \qquad (\text{Explicit})$$

<u>How do the solutions look like?</u>

If we are given an initial condition $y(x_0) = y_0$, then we get

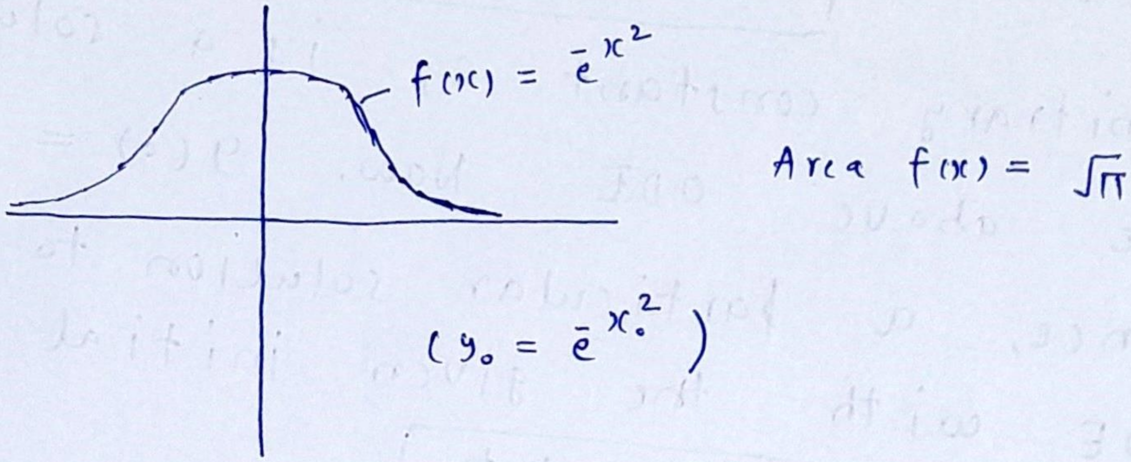
$$y_0 = c\,e^{-x_0^2} \quad \text{or} \quad c = y_0\,e^{x_0^2}$$

and from (⊛ ⊛)

$$\boxed{y(x) = y_0\,e^{(x_0^2 - x^2)}}$$

In particular, if we choose $y_0 = e^{-x_0^2}$, then

$y(x) = e^{-x^2}$ and the graph of solution (
is



Note: $y(x) = 0$ is a solution of the original equation (*) which was lost in the separation process.

Example: Given an amount of a radioactive substance, say 1 gm, find the amount present at any later time.

As we have already seen, the relevent ODE is $\frac{dy}{dt} = -ky$.

The initial amount given is 1 gm at time $t = 0$, i.e. $y(0) = 1$.

By the method of separation of variables, we have $\frac{dy}{y} = -k\,dt$, $y(x) \neq 0$.

ntegration of the above leads to

$$\ln y = -kt + \ln c.$$

Therefore, $\boxed{y(x) = ce^{-kt}}$, for an ⑥

arbitrary constant c, is a solution of the above ODE. Now, $y(0) = 1 \Rightarrow c = 1$.

Hence, a particular solution to the above ODE with the given initial condition is $\boxed{y(x) = e^{-kt}.}$

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Lecture - 5 & 6

Definition (solution of an ODE of order

A function $\phi(x)$ is said to be a solution of an n-th order ODE

$$y^{(n)} = f(x, y, y', \dots, y^{n-1}) \quad \text{——} \quad (*)$$

in an interval $\alpha < x < \beta$ if ϕ has atleast n derivatives : $\phi', \phi'', \dots, \phi^{(n)}$

and $$\phi^{(n)}(x) = f(x, \phi(x), \phi'(x), \dots, \phi^{(n-1)}(x))$$

$$\alpha < x < \beta.$$

Remarks : ① The interval $\alpha < x < \beta$ is known as the interval of existence or interval of definition for the solution.

② A solution to a DE is also known as an integral of the equation and it's graph is called a solution curve or an integral curve.

Explicit and Implicit solution:

An explicit solution of the n-th order ODE is a solution (real valued) in which the dependent variable is expressed in terms of the independent variable and constants.

That is, an explicit solution is a solution $y(x)$ of the form $y = \phi(x)$ on $\alpha < x < \beta$.

Implicit solution: A relation $g(x, y) =$ is called an implicit solution of (*) if this relation defines atleast one function, $\phi(x)$ in an interval $\alpha < x < \beta$ such that the function is an explicit solution of (*) in this interval.

Example: Consider $x + yy' = 0$.

$x^2 + y^2 - 25 = 0$ is an implicit solution of $x + yy' = 0$ in $-5 < x < 5$ because it defines two functions

$$\phi_1(x) = \sqrt{25 - x^2}, \quad \phi_2(x) = -\sqrt{25 - x^2},$$

which are explicit solutions of the DE in the given interval. (Verify)

Exercise: Show that the function $\phi(x) = x^2 - x^{-1}$ is an explicit solution to the ODE

$$\frac{d^2y}{dx^2} - \frac{2}{x^2}y = 0. \quad \text{——} \quad (a)$$

Hint: Substitute $\phi(x)$ in equation (a) and see if it satisfies the differential equation.

Formal solution: A relation $g(x,y) = 0$ is called a formal solution of

$$y^{(n)} = f(x, y, y', \dots, y^{(n-1)}),$$

if on differentiating the former (n-times) w.r.t. x, we get back the latter.

Example: Consider $x^2 + y^2 + 25 = 0$

$$\Rightarrow \quad x + yy' = 0 \quad \Rightarrow \quad \boxed{y' = -x/y}$$

We say $x^2 + y^2 + 25 = 0$ formally satisfies $x + yy' = 0$. But it is not an implicit solution of DE as this relation does not yield 'ϕ' which is an explicit solution of the DE on any real interval I.

If we try to write

$$\boxed{y = \pm\sqrt{-25 - x^2}}$$

Then $y(x)$ is not a solution for the differential equation in any interval.

Homogeneous equation:

Definition: A function $f(x_1, x_2, \ldots, x_n)$ is called homogeneous of degree 'd' if

$$f(tx_1, tx_2, \ldots, tx_n) = t^d f(x_1, \ldots, x_n)$$

for some $d \in \mathbb{Z}$ and for any scalar t.

The number $d \in \mathbb{Z}$ is called the degree of $f(x_1, \ldots, x_n)$.

Example: $f(x,y) = x^2 + xy + y^2$ is homogeneous of degree 2.

$f(x,y) = y + x\cos^2\left(\frac{y}{x}\right)$ is homogeneous of degree 1.

Definition: The first order ODE

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

is called homogeneous if M and N are homogeneous functions of degree d.

Example: $(y^2 - x^2)\frac{dy}{dx} + 2xy = 0.$

Let $M(x,y) + N(x,y)\frac{dy}{dx} = 0$, where M and N are homogeneous of degree d.

Put $\frac{y}{x} = v$. Then, $\frac{dy}{dx} = x\frac{dv}{d} + v.$

Substituting this in the given ODE,

we get

$$M(x, vx) + N(x, vx)\left(x\frac{dv}{dx} + v\right) = 0.$$

Thus,

$$x^d M(1, v) + x^d N(1, v)\left(x\frac{dv}{dx} + v\right) = 0.$$

Let $x \neq 0$. Then

$$M(1, v) + N(1, v)\, v + N(1, v)\, x\frac{dv}{dx} = 0.$$

Thus,

$$\frac{dx}{x} + \frac{N(1, v)}{M(1, v) + N(1, v)\cdot v}\, dv = 0.$$

This is a separable equation.

<u>Exercise</u>: Let $\frac{dy}{dx} = f(x, y)$, where $f(x, y)$ is homogeneous of degree '0'. Then show that it can be reduced to variable separable DE.

<u>Hint</u>: Substitute $y = xv$ and obtain

$$\frac{dx}{x} = \frac{dv}{f(1, v) - v}.$$

<u>Example</u>: Solve the ODE

$$(y^2 - x^2)\frac{dy}{dx} + 2xy = 0.$$

Sol^n: We can write

$$\frac{dy}{dx} = \frac{2xy}{x^2 - y^2}.$$

The RHS is homogeneous of degree '0'.

Hence $$\frac{dx}{x} = \frac{dv}{\frac{2v}{1-v^2} - v} = \frac{(1-v^2)\,dv}{v + v^3}$$

On integrating, we get

$$\ln|x| = \int \left[\frac{1}{v} - \left(\frac{2v}{v^2+1}\right)\right] dv + c_1.$$

or $$\ln|x| + \ln(v^2+1) - \ln|v| = c_1.$$

Hence, $$\frac{x(v^2+1)}{v} = 2c,$$

or $$y^2 + x^2 = 2cy$$

$$\Rightarrow \boxed{x^2 + (y-c)^2 = c^2}$$

Problem: Solve the following differential equation

$$\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2}$$

Sol^n: Note that $f(x,y) = \dfrac{x^2 + xy + y^2}{x^2}$ is a homogeneous function of degree '0'.

As discussed before, the substitution $\boxed{y = vx}$ leads to $$\frac{dv}{dx} = \frac{f(1,v) - v}{x}.$$

Now $f(1,v) = \frac{1+v+v^2}{1} = 1+v+v^2$

$$\frac{dv}{dx} = \frac{1+v+v^2-v}{x} = \frac{1+v^2}{x}$$

$$\Rightarrow \frac{1}{x}\,dx = \frac{1}{1+v^2}\,dv$$

$$\Rightarrow \tan^{-1} v = \ln|x| + c$$

$$\Rightarrow \tan^{-1}(y/x) - \ln|x| = c$$

for some constant c, is the general solution of the ODE.

Problem: Sove the equation $x^3 (\sin y)\, y' = 2$. Find a particular solution such that $y(x) \to \pi/2$ as $x \to \infty$.

Sol^n: Given $x^3 (\sin y) \frac{dy}{dx} = 2$

$$\Rightarrow \sin y \frac{dy}{dx} = \frac{2}{x^3}$$

$$\sin y \, dy = \frac{2}{x^3}\,dx \quad \text{———} \quad (*)$$

Integration of $(*)$ leads to

$$-\cos y + c = -\frac{1}{x^2}$$

$\frac{1}{x^2} - \cos y + c = 0$. This is an implicit solution of the given DE.

Given that $x \to \infty$, $y(x) \to \pi/2$. (8)

Note that as $x \to \infty$, $\frac{1}{x^2} \to 0$

$$\left.\begin{array}{l} \frac{1}{x^2} \to 0 \\ y(x) \to \frac{\pi}{2}, \quad \cos y \to 0 \end{array}\right\} (**)$$

Substituting $(**)$ in $(*)$, we obtain

$\boxed{C = 0}$. Therefore the particular solution of given ODE is $\boxed{\frac{1}{x^2} - \cos y = 0.}$

Problem: Solve the differential equation

$y'(x) = f(x)$ for $x \in \mathbb{R}$ where

$$f(x) = \begin{cases} e^x & \text{if } x \geq 2 \\ e^2 & \text{if } x < 2. \end{cases}$$

Sol^n: Note that the right hand side function $f(x)$ is continuous for $x \in \mathbb{R}$. Therefore by the first fundamental theorem of integral calculus $y(x)$ is differentiable function.

$$\frac{dy}{dx} = e^2 \quad \text{if } x < 2.$$

$$\boxed{y(x) = e^2 x + C_1, \quad x < 2} \quad \text{—— (a)}$$

since $y(x)$ is continuous function

$$\boxed{\lim_{x \to 2^-} y(x) = y(2)} \quad \text{—— } (*)$$

For for $x \geq 2$, $\frac{dy}{dx} = e^x$.

$$\boxed{y(x) = e^x + c_2} \quad \text{— (b)}$$

Using (*) in (a) and (b), we get

$$e^2 + c_2 = e^2 \cdot 2 + c_1$$

$$\Rightarrow \quad c_2 - c_1 = e^2 \quad \Rightarrow \quad \boxed{c_2 = c_1 + e^2} \quad \text{— (c)}$$

From (a), (b) and (c), we have

$$y(x) = \begin{cases} e^2 x + c_1 & x < 2 \\ e^x + e^2 + c_1 & x \geq 2 \end{cases}$$

* choose $c_1 = 1$, then $c_2 = 1 + e^2$

and $$y(x) = \begin{cases} e^2 x + 1 & x < 2, \\ e^x + 1 + e^2 & x \geq 2. \end{cases}$$